

Use-Case Flow Specification

Use Case: UC-03 — Reserve Equipment Slot

Primary Actor: Student

1. Preconditions

- Student is authenticated.
- The selected equipment exists in the system.
- The equipment has a valid calibration status.
- The requested reservation date is within 7 days.
- The requested reservation duration does not exceed 2 hours.

2. Main Success Scenario

1. Student logs into the Smart Lab Equipment & Slot Reservation Portal.
2. Student selects the required laboratory equipment.
3. System displays the equipment's availability and calibration status.
4. Student selects an available date and time slot.
5. System verifies that the requested slot is within the allowed 7-day reservation window.
6. System verifies that the reservation duration does not exceed 2 hours.
7. System checks that the equipment is calibrated.
8. System locks the requested slot.
9. System creates the reservation.
10. System generates a reservation token.
11. System displays the reservation confirmation to the student.
12. Use case ends successfully.

3. Alternate Flow — Selected Slot Becomes Unavailable

- 8a1.** At Step 8, the system detects that another reservation has locked the requested slot.
- 8a2.** System rejects the reservation request.
- 8a3.** System informs the student that the selected slot is unavailable.
- 8a4.** System displays other available slots for the equipment.
- 8a5.** Student selects another available slot.
- 8a6.** System processes the new reservation request.
- 8a7.** If no suitable slot is selected, the use case ends without creating a reservation.

4. Postconditions

Success: Reservation is stored, the selected slot is no longer available to other students, and a reservation token is generated.

Failure: No reservation is created for the unavailable slot.